# Mitigating Odor and Greenhouse Gas Emissions in Composting: Effect on Microbial additive

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## 1. Introduction

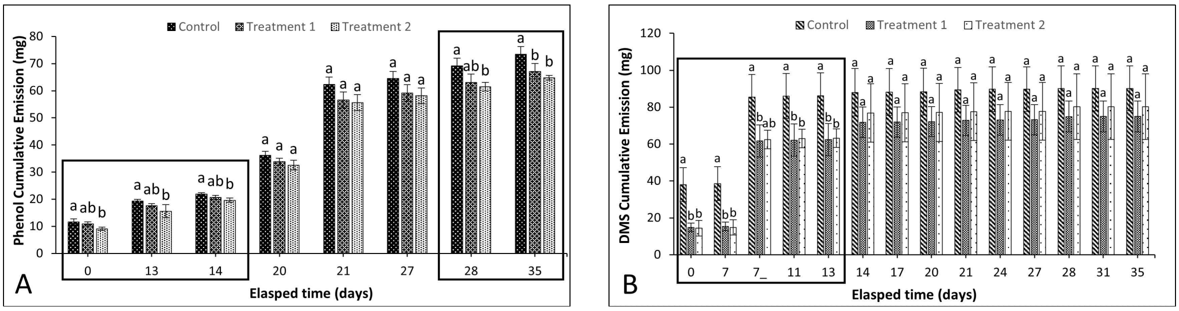
The global rise in population has led to a significant surge in meat consumption, resulting in increased meat production and, consequently, higher volumes of livestock manure. This growth raises environmental concerns, with potential pollution such as odorous and greenhouse gases (Feilberg et al., 2017; OECD-FAO Agricultural Outlook 2022-2031). Composting offers a sustainable solution by transforming livestock manure into nutrient-rich organic fertilizers. Optimizing the composting process involves the introduction of microbial cultures, including bacteria, fungi, and actinomycetes, which enhance decomposition, reduce odorous compounds, and improve compost quality (Rastogi et al., 2020). Understanding the role of microbial additives is pivotal for enhancing composting efficiency and compost quality. Therefore, this research aims to determine the mitigation effect of microbial addition on ammonia, greenhouse gases, volatile fatty acids (VFAs), volatile organic compounds (VOCs), and sulfur compounds during cattle and swine manure composting.

## 2. Methods

The experiment was conducted with laboratory-scale reactors (3.8L) for 35 days using *Aquamicrobium lusatiense* NLF as a microbial additive. The mix ratio of swine manure, dairy cattle manure, and sawdust was 5:4:1. Treatment-1 was mixed with 1% microbial cultures on the initial day, while Treatment-2 was conducted with 0.5% microbial addition on the initial day and 0.5% on the second week. The compost matrix was aerated at 0.4 L/min-VS during the first week and 0.3 L/min-VS after the first week. Mitigation effects were determined by measuring odorous gases, including $NH_3$ (ammonia), sulfur compounds involving $H_2S$ (hydrogen sulfide), $CH_4S$ (methyl mercaptan), $C_2H_6S$ (dimethyl sulfide), and $C_2H_6S_2$ (dimethyl disulfide), as well as VFAs such as acetic acid, propionic acid, iso-butyric acid, butyric acid, iso-valeric acid, and valeric acid, then VOCs including phenol, p-Cresol (4-Methylphenol), indole, and skatole (3-Methylindole).

## 3. Results and Discussion

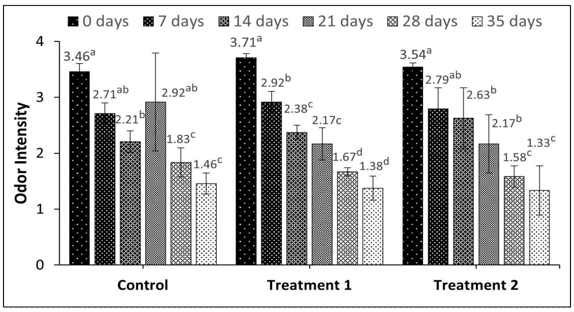
In this study, phenol emissions as a significant VOC were closely monitored (Figure 1A). After two weeks of composting, a significant difference was observed, with Treatment-2 demonstrating a 10.5% reduction in phenol emissions compared to the control ($p<0.05$). This finding suggests that Treatment-2 has a notable impact on reducing phenol emissions early in the composting process. However, the data also showed that between days 20 and 27, there was no significant difference in phenol emissions ($p>0.05$). As the composting process continued, a significant difference emerged after 28 days, with Treatment-2 continuing to outperform the control in reducing phenol emissions. By the end of composting, Treatment-1 and Treatment-2 exhibited substantial reductions in phenol cumulative emissions compared to the control, with Treatment-2 leading with an 11.7% reduction and Treatment-1 8.6%. These results suggest that the treatments have a cumulative effect on reducing phenol emissions throughout the composting process, and their potential to improve the environmental quality of the compost.




*Different superscripts on the same days meaning each group is significantly different (p<0.05), *7_: Day 7 after mixing
Fig. 1. Phenol (A) and DMS (B) emissions during composting

In the case of DMS, the initial 13 days of composting showed a remarkable reduction in both treatments compared to the control ($p<0.05$) (Figure 1B). Treatment-1 exhibited a substantial 27.6% reduction, while Treatment-2 achieved a 26.7% reduction. These findings signify the effectiveness of these treatments in mitigating DMS emissions early in the composting process. As the composting process continued, it is noteworthy that no significant differences were observed in cumulative DMS emissions after day 14 until the end of composting ($p>0.05$). By the end of the 35-day period, Treatment-1 achieved a 16.9% reduction, and Treatment-2 achieved an 11.0% reduction. The study by Jeong et al. (2023) supports these results, *Aquamicrobium lusatiense* NLF is a sulfur-oxidizing bacteria that can reduce sulfur compounds such as DMS.

The results demonstrate that several gasses, including $NH_3$, greenhouse gases, sulfur compounds ($H_2S$), and VFA (acetic acid and propionic acid), did not show a significant difference between the control group and the treatment groups ($p>0.05$) after 35 days composting. However, it's worth noting that Treatment 2 showed a slight reduction in $NH_3$ emissions by 3.2% and $CH_4$ emissions by 14.3%. The substantial reduction rates in odor intensity for both treatments when comparing the first and final days indicate their effectiveness in mitigating odors (Figure 2). Control has a reduction rate of 57.8±5.6%, while treatments 1 and 2 have as much as 63±5.5% and 62.4±12.5%, respectively. However, the odor unit showed no statistical difference between treatment and control on the same day during the composting ($p>0.05$).




*Different superscripts on the same group meaning the significantly different (p<0.05)
Fig. 2. The results of odor intensity over 35 days

## 4. Conclusion

The results indicate the potential of *Aquamicrobium lusatiense* NLF to reduce phenol and DMS emissions in the composting process, which are significant contributors to environmental and odor concerns in composting. Further research could focus on optimizing these treatments and exploring their long-term effects on compost quality and microbial communities. Considering environmental and odor-related factors, these results contribute to more sustainable composting practices.

## Acknowledgment

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